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Course & Year: BSIT-4

Capture The Flag

Mission: Learn how Angular applications are built using reusable components.

Objective: Transform your first Angular application into a modular application using reusable components while following a basic Git workflow.

Flag 0 – Return to Base

Mission

Open the Angular project you created during the previous activity.

Tasks

- Open **Visual Studio Code**.
- Open your Angular project folder.
- Open the integrated terminal.
- Verify that the project still runs.
- ng serve

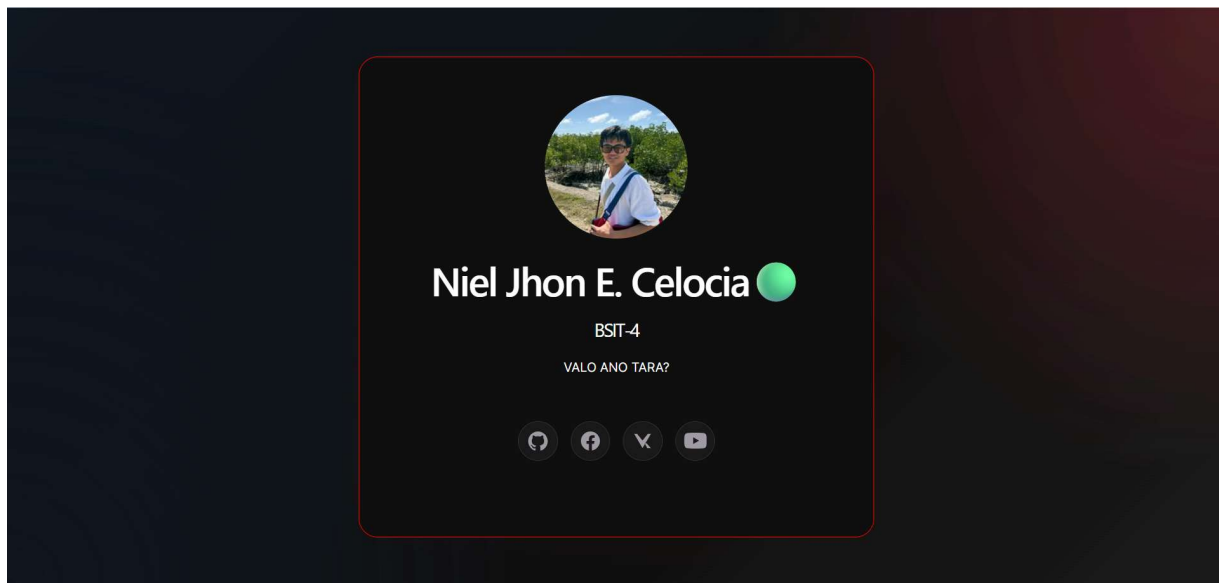
Open:

<http://localhost:4200>

If your Angular welcome page appears, proceed.

Evidence

- VS Code with project opened
- Terminal showing **ng serve**
- Browser displaying the Angular application



Flag 1 – Mission Branch

Mission

Every new feature must have its own Git branch.

Create a new branch named:

feature/components

Commands:

Evidence

- Terminal showing the successful branch creation
- Output of `git branch`

```
nielj@Ejann MINGW64 ~/my-first-angular-app (main)
$ git branch feature/component
```

```
nielj@Ejann MINGW64 ~/my-first-angular-app (main)
$ git checkout feature/components
Switched to branch 'feature/components'

nielj@Ejann MINGW64 ~/my-first-angular-app (feature/components)
$
```

Flag 2 – Explore the Base

Mission

Investigate your Angular project before making changes.

Locate the following:

- app.html
- app.ts
- app.css

- app.spec.ts

Understand that these files belong to the **App Component**, the root component of the application.

Evidence

- Explorer showing the files
- **app.html** opened

```
src > app > app.html > HTML Language Features > style > .host
Go to component
1 <!-- ***** -->
2 <!-- ***** The content below ***** -->
3 <!-- ***** is only a placeholder ***** -->
4 <!-- ***** and can be replaced. ***** -->
5 <!-- ***** -->
6 <!-- ***** Delete the template below ***** -->
7 <!-- ***** to get started with your project! ***** -->
8 <!-- ***** -->
9
10 <style>
11   :host {
12     --bright-blue: oklch(51.01% 0.274 263.83);
13     --electric-violet: oklch(53.18% 0.28 296.97);
14     --french-violet: oklch(47.66% 0.246 305.88);
15     --vivid-pink: oklch(69.02% 0.277 332.77);
16     --hot-red: oklch(61.42% 0.238 15.34);
17     --orange-red: oklch(63.32% 0.24 31.68);
18
19     --gray-900: oklch(19.37% 0.006 300.98);
20     --gray-700: oklch(36.98% 0.014 302.71);
21     --gray-400: oklch(70.9% 0.015 304.04);
22
23     --red-to-pink-to-purple-vertical-gradient: linear-gradient(180deg,
24       var(--orange-red) 0%,
25       var(--vivid-pink) 50%,
26       var(--electric-violet) 100%);
27
28     --red-to-pink-to-purple-horizontal-gradient: linear-gradient(90deg,
29       var(--orange-red) 0%,
30       var(--vivid-pink) 50%,
31       var(--electric-violet) 100%);
32
33     --pill-accent: var(--bright-blue);
34
35     font-family: "Inter", -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
36       Helvetica, Arial, sans-serif, "Apple Color Emoji", "Segoe UI Emoji",
```

Flag 3 – Clear the Battlefield

Mission

Remove the default Angular welcome page.

Delete everything inside:

app.component.html

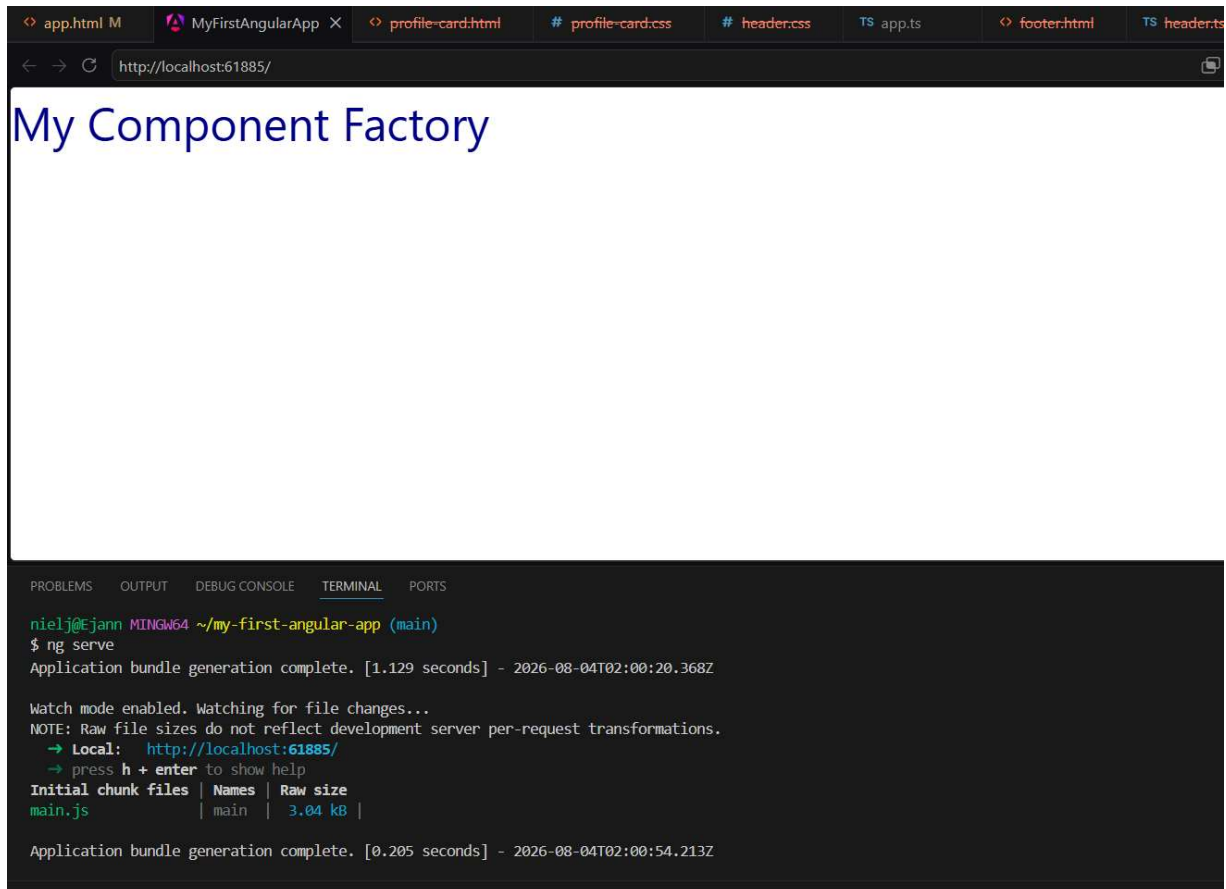
Replace it with:

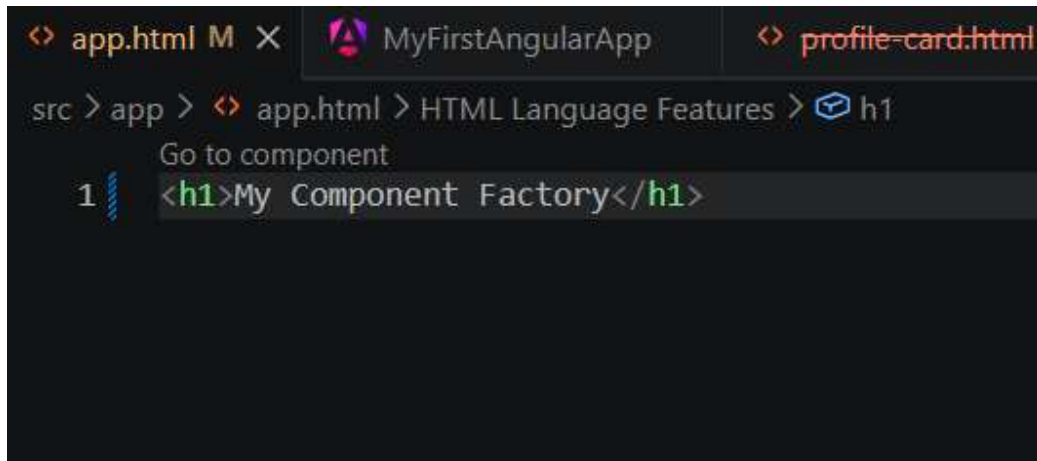
```
<h1>My Component Factory</h1>
```

Verify the browser updates.

Evidence

- Browser showing only the heading
- HTML file





Flag 4 – Build the Headquarters

Mission

Create a reusable Header component.

Generate:

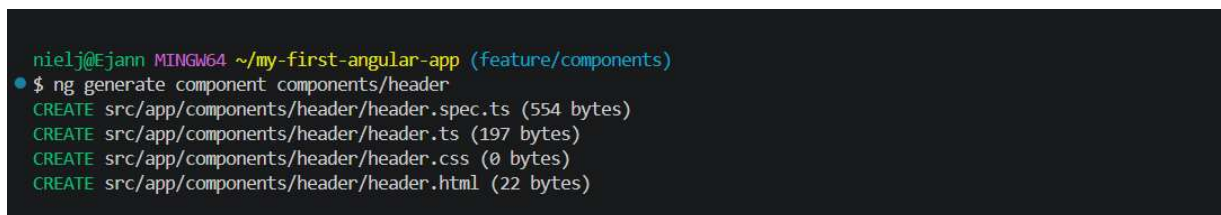
```
ng generate component components/header
```

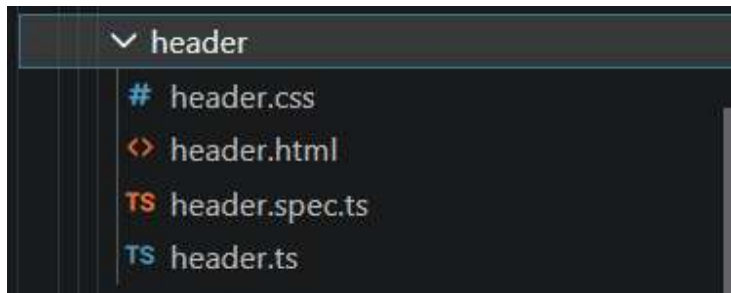
Angular should automatically generate:

header.ts
header.html
header.css
header.spec.ts

Evidence

- Terminal
- Explorer showing generated files





Flag 5 – Raise the Banner

Mission

Inside the Header component, create a simple header.

Example:

```
<header>
  <h2>Don Bosco Student Portal</h2>
</header>
```

Use the component inside:

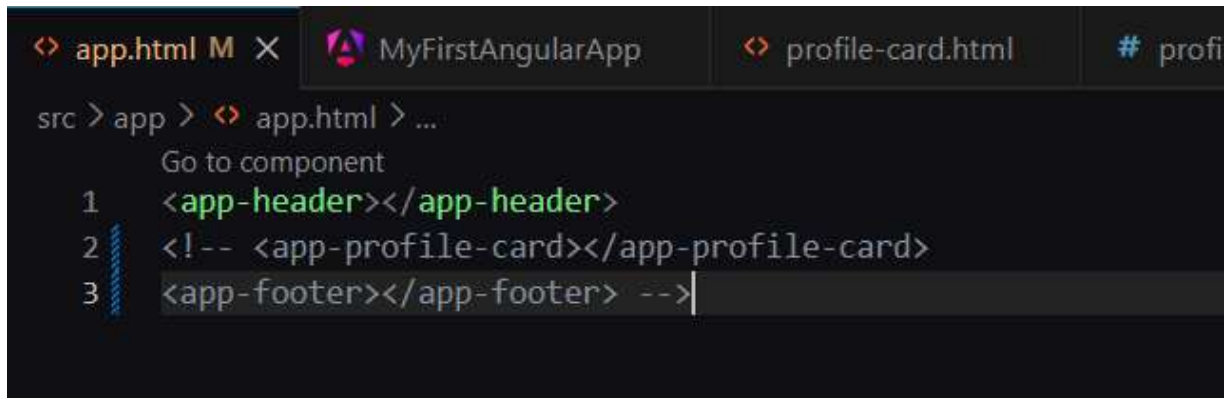
```
app.html
<app-header></app-header>
```

Verify it appears in the browser.

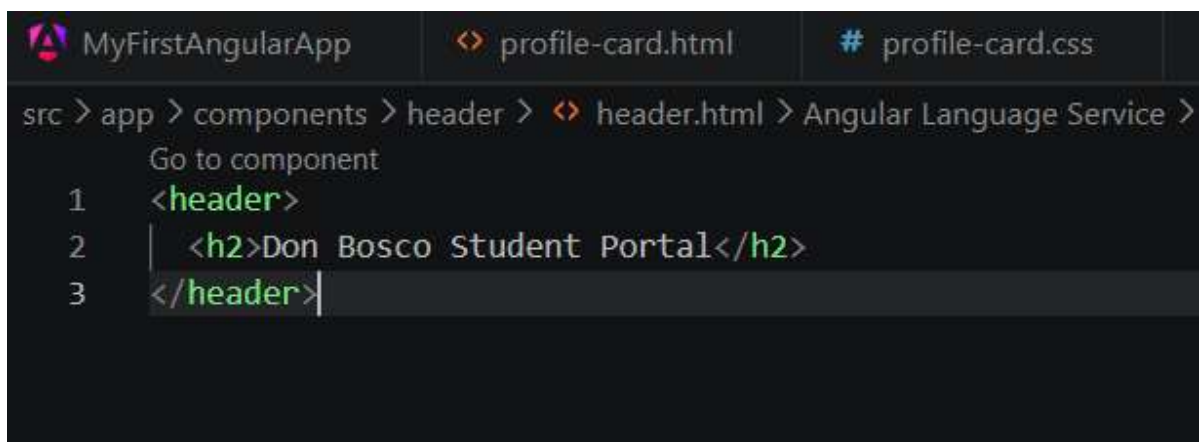
Evidence

- Browser
- HTML code





```
src > app > app.html > ...  
Go to component  
1 <app-header></app-header>  
2 <!-- <app-profile-card></app-profile-card>  
3 <app-footer></app-footer> -->
```



```
src > app > components > header > header.html > Angular Language Service >  
Go to component  
1 <header>  
2 | <h2>Don Bosco Student Portal</h2>  
3 </header>
```


Flag 6 – Secure the Exit

Mission

Create a Footer component.

Use the same command previously, but now for the footer.

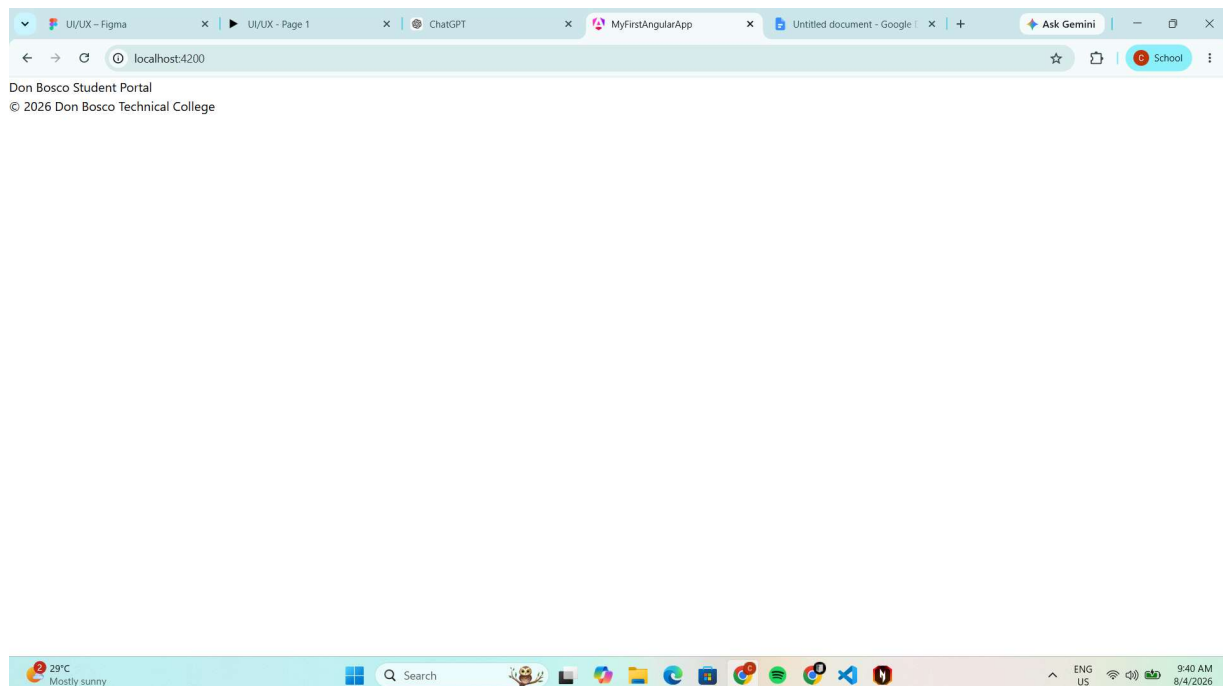
Display:

```
<footer>  
  © 2026 Don Bosco Technical College  
</footer>
```

Insert it into the App Component.

Evidence

- Browser
- Generated files



```
nielj@Ejann MINGW64 ~/my-first-angular-app (feature/components)
• $ ng g c components/footer
CREATE src/app/components/footer/footer.spec.ts (554 bytes)
CREATE src/app/components/footer/footer.ts (197 bytes)
CREATE src/app/components/footer/footer.css (0 bytes)
CREATE src/app/components/footer/footer.html (22 bytes)

nielj@Ejann MINGW64 ~/my-first-angular-app (feature/components)
• $ git add .
```

Flag 7 – Create the Profile Card

Mission

Generate a reusable Profile Card.

Use the command to create a new component called profile-card.

Create a card containing:

- Profile Picture
- Name
- Course
- Year Level

(Design is optional)

Insert it into:

app.html

Evidence

- Browser
- Profile Card component

Don Bosco Student Portal



© 2026 Don Bosco Technical College

```
nielj@Ejann MINGW64 ~/my-first-angular-app (feature/components)
$ ng g c components/profile-card
CREATE src/app/components/profile-card/profile-card.spec.ts (590 bytes)
CREATE src/app/components/profile-card/profile-card.ts (220 bytes)
CREATE src/app/components/profile-card/profile-card.css (0 bytes)
CREATE src/app/components/profile-card/profile-card.html (28 bytes)
```

Flag 8 – Produce Announcement Cards

Mission

Generate one reusable Announcement Card.

Use the same command, creating the component announcement-card.

Design a card containing:

- Announcement Title
- Description
- Date

Do **NOT** create three separate components.

Instead, reuse the same component.

Example:

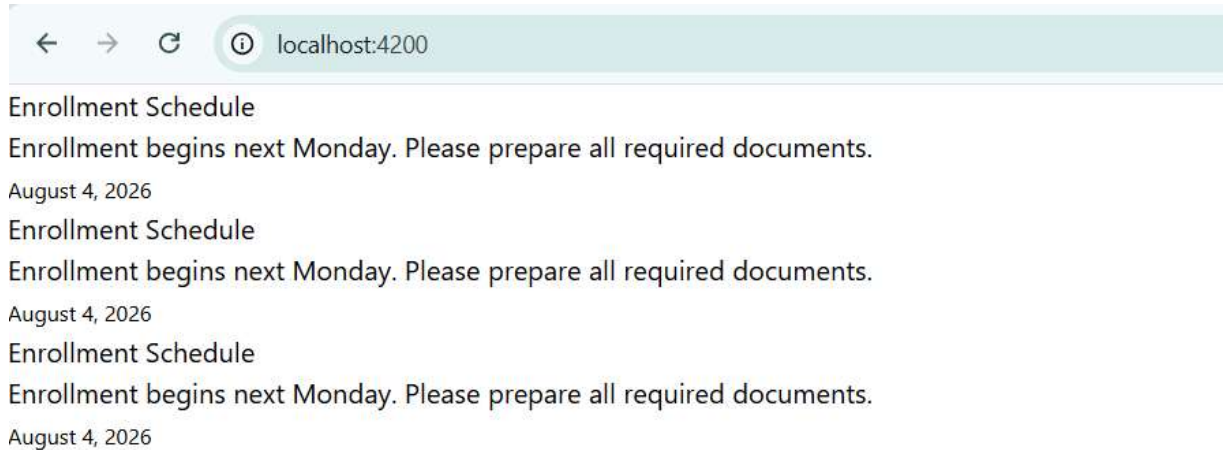
```
<app-announcement-card></app-announcement-card>
```

```
<app-announcement-card></app-announcement-card>
```

```
<app-announcement-card></app-announcement-card>
```

Evidence

- Browser
- App component showing three announcement cards



Flag 9 – Assemble the Dashboard

Your `app.html` should now act only as the page assembler.

Assemble your page to have this structure:

header

profile-card

announcement-card

announcement-card

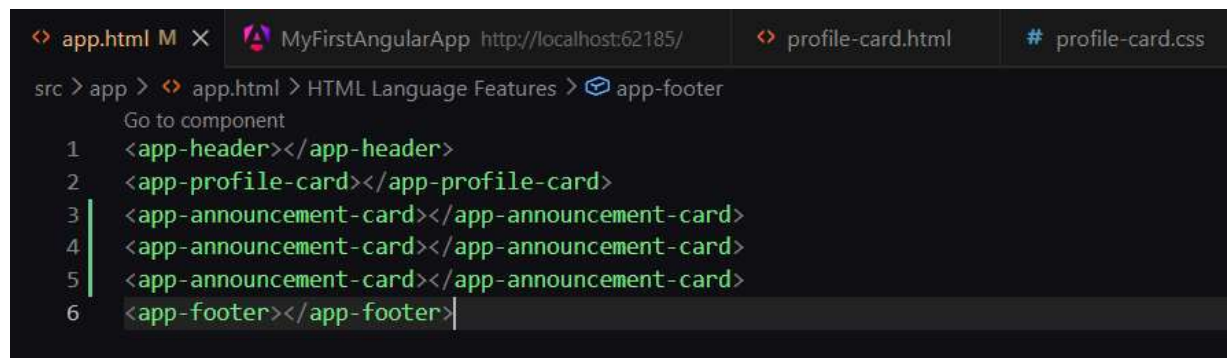
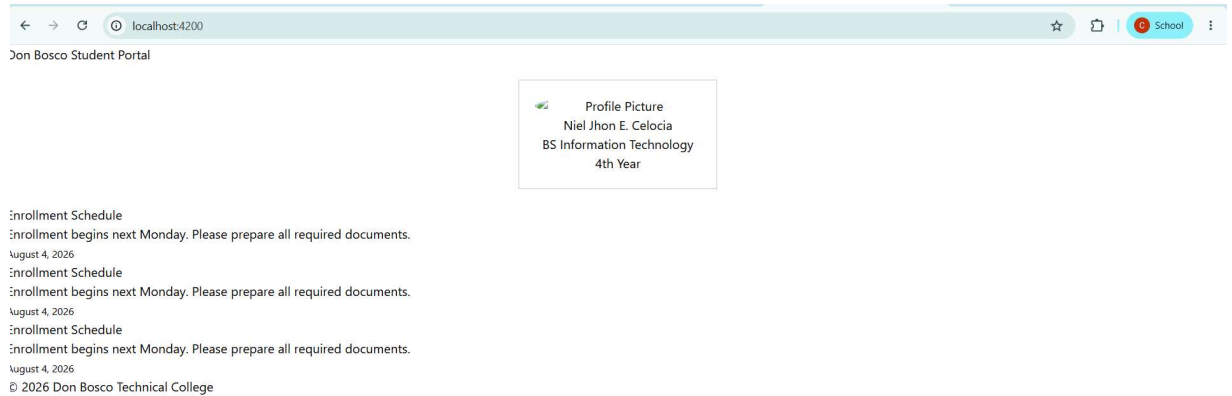
announcement-card

footer

Notice that almost no actual page content exists here. Only reusable components.

Evidence

- Browser
- `app.html`



Flag 10 – Client Changed Their Mind

Mission

The client sends an urgent request:

"Every announcement card should now have:

- Rounded corners
- A blue border
- Padding

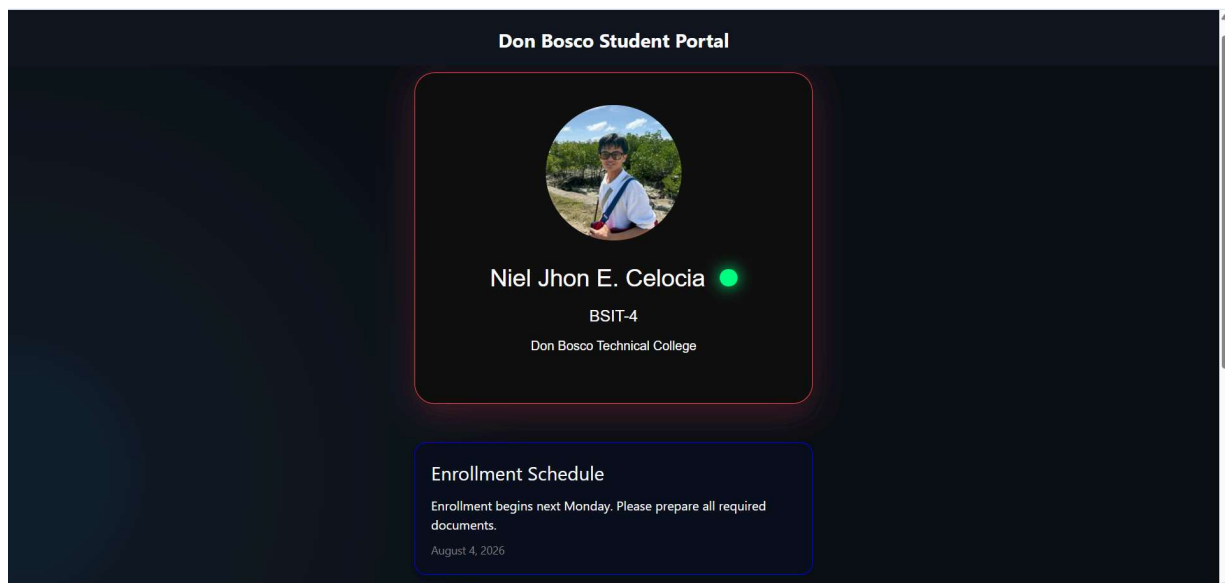
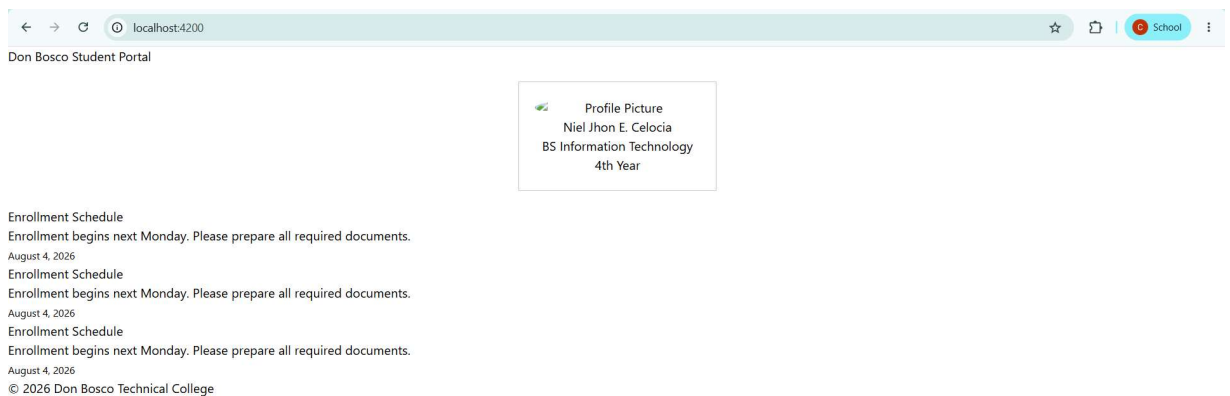
- A subtle shadow"

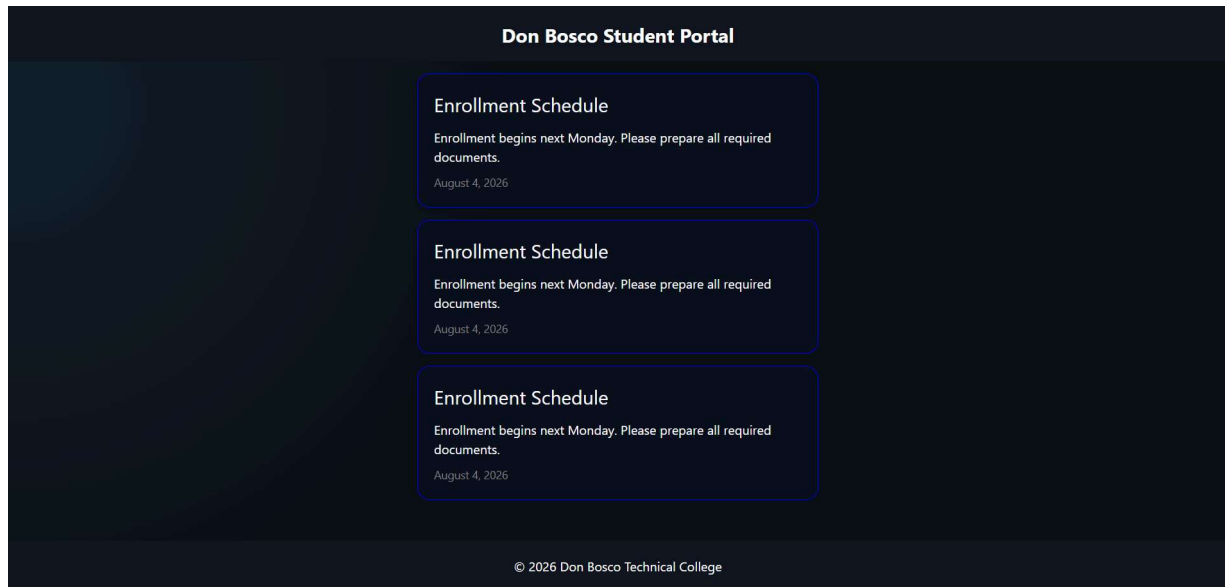
You must update **only one file**.

Apply your styles there and verify that **all announcement cards** update automatically.

Evidence

- Browser before and after the CSS change
- `Announcement-card.css` code





```

src > app > components > announcement-card > # announcement-card.css > announcement-card
1  .announcement-card{
2
3      border: 1px solid blue;
4
5      border-radius:15px;
6
7      padding:20px;
8
9      margin:15px auto;
10
11     width:500px;
12
13     background: #001120;
14
15     box-shadow:0 4px 10px rgba(0,0,0,.15);
16
17     display:flex;
18     flex-direction:column;
19
20 }
21
22
23
24 .announcement-card h3{
25
26     color:rgb(253, 254, 255);
27
28     font-size:24px;
29
30     margin-bottom:10px;
31
32 }
33
34 .announcement-card p{
35
36     font-size:16px;
37
38     line-height:1.5;
39
40     margin-bottom:10px;
41
42     color:white;
43
44 }
45
46 .announcement-card small{
47
48     color:gray;
49
50     font-size:14px;
51
52 }
53
54 .announcement-card:hover{
55
56     h3{

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

niel@Ejann MINGW64 ~/my-first-angular-app (feature/components)
$ ng serve
Initial chunk files | Names | Raw size
main.js | main | 19.85 kB |
Application bundle generation complete. [0.115 seconds] - 2026-08-04T03:21:02.870Z
Component update sent to client(s).

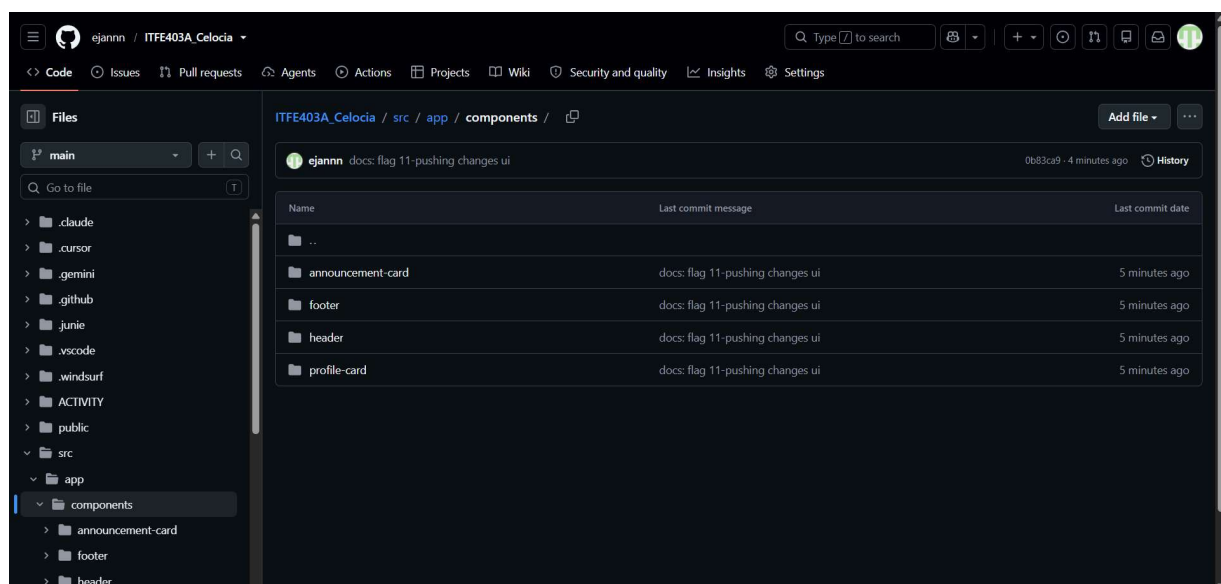
```


Flag 11 – Pushing Changes

Commit your completed work, then push to your repository.

Evidence

- Terminal commands for committing + pushing
- Screenshot of the Github Repository reflecting the pushed changes



```

nielj@Ejann MINGW64 ~/my-first-angular-app (main)
$ git merge feature/components
src/app/app.ts | 8 +-
.../announcement-card/announcement-card.css | 77 +++++
.../announcement-card/announcement-card.html | 10 +
.../announcement-card/announcement-card.spec.ts | 22 ++
.../announcement-card/announcement-card.ts | 9 +
src/app/components/footer/footer.css | 13 +
src/app/components/footer/footer.html | 3 +
src/app/components/footer/footer.spec.ts | 22 ++
src/app/components/footer/footer.ts | 9 +
src/app/components/header/header.css | 17 ++
src/app/components/header/header.html | 3 +
src/app/components/header/header.spec.ts | 22 ++
src/app/components/header/header.ts | 10 +
src/app/components/profile-card/profile-card.css | 98 ++++++
src/app/components/profile-card/profile-card.html | 21 ++
.../components/profile-card/profile-card.spec.ts | 22 ++
src/app/components/profile-card/profile-card.ts | 9 +
19 files changed, 398 insertions(+), 342 deletions(-)
create mode 100644 src/app/components/announcement-card/announcement-card.css
create mode 100644 src/app/components/announcement-card/announcement-card.html
create mode 100644 src/app/components/announcement-card/announcement-card.spec.ts
create mode 100644 src/app/components/announcement-card/announcement-card.ts
create mode 100644 src/app/components/footer/footer.css
create mode 100644 src/app/components/footer/footer.html
create mode 100644 src/app/components/footer/footer.spec.ts
create mode 100644 src/app/components/footer/footer.ts
create mode 100644 src/app/components/header/header.css
create mode 100644 src/app/components/header/header.html
create mode 100644 src/app/components/header/header.spec.ts
create mode 100644 src/app/components/header/header.ts
create mode 100644 src/app/components/profile-card/profile-card.css
create mode 100644 src/app/components/profile-card/profile-card.html
create mode 100644 src/app/components/profile-card/profile-card.spec.ts
create mode 100644 src/app/components/profile-card/profile-card.ts

nielj@Ejann MINGW64 ~/my-first-angular-app (main)
$ git push origin main
Enumerating objects: 92, done.
Counting objects: 100% (92/92), done.
Delta compression using up to 12 threads
Compressing objects: 100% (79/79), done.
Writing objects: 100% (86/86), 9.82 KiB | 558.00 KiB/s, done.
Total 86 (delta 34), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (34/34), completed with 2 local objects.
To https://github.com/ejannn/ITFE403A_Celocia.git
f4b747c..0b83ca9 main -> main

nielj@Ejann MINGW64 ~/my-first-angular-app (main)

```

